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Preparing abstract and title

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Academic Ethics and Publication
Faculty of Engineering
Universitas Gadjah Mada

2025


ABSTRACT
PIZZA
HOT & DELICIOUS
PAPER


Dr. Geistbot, PhD
@Geistb0t
the abstract
the paper


My paper's title
@phddarymemes
My paper's abstract
The rest of the paper

Abstract

A 'mini' version of the paper, but only highlight the important findings

Most journal has word limitation, around 80-250 words → 1 paragraph

The most read part of a journal article → a good abstract enhance the study's visibility (likely leads to increase citation)

Usually written last (despite appearing at the beginning of the paper)

Structure:

- A strong research statement/problem → *first sentence leaves strong impression*, provide introduction or context to strengthen the statement.
- Summary of approach/method(s)
- Key results/findings
- Conclusions
- Implications → e.g., future research direction, policy insight

"The growing economic resemblance of spouses has contributed to rising inequality by increasing the number of couples in which there are two high- or two low-earning partners. The dominant explanation for this trend is increased assortative mating. Previous research has primarily relied on cross-sectional data and thus has been unable to disentangle changes in assortative mating from changes in the division of spouses' paid labor—a potentially key mechanism given the dramatic rise in wives' labor supply. We use data from the Panel Study of Income Dynamics (PSID) to decompose the increase in the correlation between spouses' earnings and its contribution to inequality between 1970 and 2013 into parts due to (a) changes in assortative mating, and (b) changes in the division of paid labor. Contrary to what has often been assumed, the rise of economic homogamy and its contribution to inequality is largely attributable to changes in the division of paid labor rather than changes in sorting on earnings or earnings potential. Our findings indicate that the rise of economic homogamy cannot be explained by hypotheses centered on meeting and matching opportunities, and they show where in this process inequality is generated and where it is not." (p. 985)

The first sentence introduces the **topic** under study (the "economic resemblance of spouses"). This sentence also implies the **question** underlying this research study: what are the various causes—and the interrelationships among them—for this trend?

These next two sentences explain what **previous research** has demonstrated. By pointing out the limitations in the methods that were used in previous studies, they also provide a **rationale** for new research.

The data, research and analytical **methods** used in this new study.

The major **findings** from and **implications** and **significance** of this study.

Source: https://writing.wisc.edu/handbook/assignments/writing-an-abstract-for-your-research-paper/

Writing the abstract

- Explicitly identify your (research) contribution
  - Use ***past simple*** or ***present perfect*** instead of present simple; compare:
    - Analysis shows concordance for 74% of mutation calls → *unclear who carried out the analysis*
    - Analysis showed concordance for 74% of mutation calls → *it was the authors who carried out the analysis*
  - Use the active (we) rather than the passive.
  - To start your contribution: *It is known that / It has been previously demonstrated that* OR *In this study / Here…..*
- Avoid ambiguity: use the exact term (e.g., the XXXX method) instead of *it*, *this*, or *which*
- Use consistent terminologies (e.g., if you use ***scheme***, do NOT change it into ***framework***)
- Use a strong, reader-friendly narrative
- Keep sentence length short
- Language should cater for 'universal' reader, use acronym sparsely
- Make it impossible for the reader not to understand the Abstract AND identify your contribution.

Avoid these:

- Cut and paste from your text → please paraphrase concisely
- The use of abbreviation
- Quotations → except if you are citing specific theory or method, in which the paper is about
- Generalized statement
- Ambiguity

Transportation
DOI 10.1007/s11116-012-9388-7

Justice in transport as justice in accessibility: applying Walzer's 'Spheres of Justice' to the transport sector

Karel Martens

Abstract This paper seeks to provide a theoretical basis for a distributive approach to transport. Using the theory developed by Michael Walzer in his 'Spheres of Justice' (1983), I argue that the transport good, defined as accessibility, should be distributed in a so-called separate sphere, i.e. independent from the way in which other key goods, like money or power, are allocated. I subsequently explore what kind of justice principle could guide the distribution of the transport good, once a separate sphere would be established. This preliminary exploration results in the elimination of a number of widely supported distributive principles, and in the tentative identification of a criterion matching the particularities of the transport good. The explorations in the paper are not intended as final answers, but rather seek to open the debate about the need for an explicit distributive transport policy and the distributive principle that should guide such a policy.

Keywords Transport · Justice · Equity · Spheres of Justice · Distribution

Examples of appropriate referencing in an abstract → mostly theoretical or method, providing a major advancement (or highly contradict) previously established work

Types of abstract

1. Simple/standard abstracts
2. Structured abstracts (have headings)
3. Abstracts with a *significance statement* or a *list of highlights*
4. Graphical abstracts

This depends on the journal style and preference

Standard abstract

- Read this abstract
- Identify components/elements on the abstract (in bold)

3 Mobilizing salt: magma-salt interactions[aa]

Abstract

Salt sequences **form** an integral part of **many** sedimentary basins **worldwide**. **Many** of these basins have experienced igneous activity either syn- or post-deposition of the salt sequences. **Despite this, little work has so far been undertaken to understand** magma-salt interactions within the subsurface, and how aspects such as salt halokinesis may be influenced by igneous activity. **Within this paper, we detail the first direct description of** relationships and textures that are developed during intrusive igneous-salt interaction. **We show that** salt composition **appears to play a dominant role** in controlling where igneous intrusions invade laterally through salt sequences in a sedimentary basin. **In particular, we illustrate that** hydrous salts, such as carnallite, **act** as preferential horizons for lateral magma intrusion. This lithological control **appears primarily related to** the heating and subsequent dehydration reaction of carnallite, which causes the carnallite to behave as viscous fluidal horizons, resulting in the non-brittle emplacement of magma, and spectacular peperitic salt-magma mingling textures. **We suggest that** heating and transformation of carnallite and other hydrous salts into viscous fluidal horizons during igneous intrusion within a regional salt sequence **may act as a possible trigger for** contemporaneous halokinesis, by creating fluid-like viscous detachment layers. Over longer time scales, however, a solidified rigid boxwork of dikes and sills **may create** zones of increased mechanical strength that will locally inhibit further salt flow.

Standard abstract

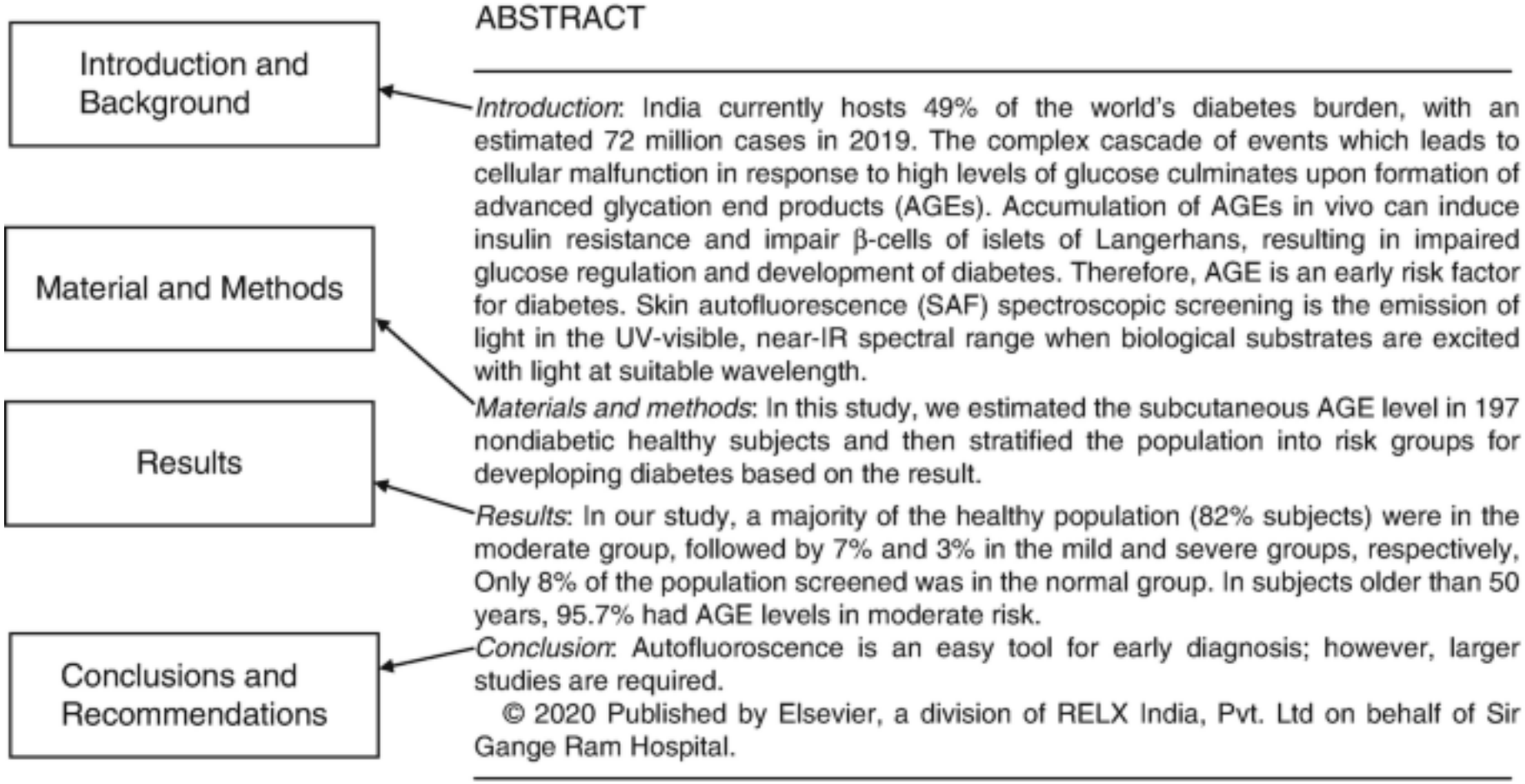
- Read this abstract
- Identify components/elements on the abstract (in bold)
- Compare with the components/elements found in the previous slide

10 Global anisotropic 3D FWI[gg]

Abstract

Seismic anisotropy **influences** both the kinematics and dynamics of seismic waveforms. **If** anisotropy **is not adequately taken into account** during full-waveform seismic inversion (FWI), then **inadequacies** in the anisotropy model **are likely to manifest as significant error** in the recovered P-wave velocity model. **Conventionally**, anisotropic FWI **uses** either a fixed anisotropy model derived from tomography or such, **or it uses** a local inversion scheme to recover the anisotropy as part of the FWI; **both of these methods can be problematic**. **In this paper, we show that** global rather than local FWI **can be used** to recover the long-wavelength anisotropy model, and **that this can then be followed by** more-conventional local FWI to recover the detailed model. **We demonstrate this approach on** a full 3D field dataset, **and show that it avoids problems** associated to cross-talk that can bedevil local inversion schemes. **Although our method** provides a global inversion of anisotropy, **it is nonetheless affordable and practical** for 3D field data.

Structured Abstract



ABSTRACT

Introduction: India currently hosts 49% of the world's diabetes burden, with an estimated 72 million cases in 2019. The complex cascade of events which leads to cellular malfunction in response to high levels of glucose culminates upon formation of advanced glycation end products (AGEs). Accumulation of AGEs in vivo can induce insulin resistance and impair β-cells of islets of Langerhans, resulting in impaired glucose regulation and development of diabetes. Therefore, AGE is an early risk factor for diabetes. Skin autofluorescence (SAF) spectroscopic screening is the emission of light in the UV-visible, near-IR spectral range when biological substrates are excited with light at suitable wavelength.

Materials and methods: In this study, we estimated the subcutaneous AGE level in 197 nondiabetic healthy subjects and then stratified the population into risk groups for deveploping diabetes based on the result.

Results: In our study, a majority of the healthy population (82% subjects) were in the moderate group, followed by 7% and 3% in the mild and severe groups, respectively, Only 8% of the population screened was in the normal group. In subjects older than 50 years, 95.7% had AGE levels in moderate risk.

Conclusion: Autofluoroscence is an easy tool for early diagnosis; however, larger studies are required.

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Some journals may ask for a structured abstract, similar to journal sections:

- Background
- Aim
- Method
- Results
- Conclusion

Source: Nundy et al., 2021 https://link.springer.com/chapter/10.1007/978-981-16-5248-6_15

Abstract + highlights (or significance)

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- Highlights
  - Addition to an abstract
  - Key take away from the study
  - In a bullet points (usually 3-5, each bullet point consists of 1 sentence)
  - Often has *character* limit (e.g., 85 characters per bullet point in Elsevier)
  - Only appear in the web version, rarely in printed or pdf-version

Abstract + highlights (or significance)

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The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour

Highlights

► **High resolution imagery** (FIB, SEM and SIMS) of carbide inclusions in uranium metal. ► **Real time images** following the reaction of the carbide inclusions with water vapour. ► **Shown** preferential consumption of carbide over that of the bulk metal. ► Quantity of impurities in the metal **therefore seriously influence** reaction rate. ► Metal purity **must be considered when** storing uranium in air or moist conditions.

Abstract

The reaction between uranium and water vapour **has been well investigated, however discrepancies exist** between the described kinetic laws, pressure dependence of the reaction rate constant and activation energies. **Here this problem is looked at by examining** the influence of impurities in the form of carbide inclusions on the reaction. Samples of uranium containing 600 ppm carbon **were analysed** during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM) system. After water exposure, samples **were analysed using** secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD). **The results of the current study indicate that** carbide particles on the surface of uranium **readily react** with water vapour to form voluminous $UO_3 \cdot xH_2O$ growths **at rates significantly faster than** that of the metal. **The observation may also have implications for previous experimental studies** of uranium-water interactions, where the presence of differing levels of undetected carbide **may partly account for** the discrepancies observed between datasets.

Abstract + highlights (or significance)

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Amazonian functional diversity from forest canopy chemical assembly

Significance

Canopy trees are **keystone organisms** that create habitat for **an enormous array of** flora and fauna and **dominate** carbon storage in tropical forests. Determining the functional diversity of tree canopies **is, therefore, critical to understanding** how tropical forests are assembled and predicting ecosystem responses to environmental change. Across the megadiverse Andes-to-Amazon corridor of Peru, **we discovered** a large-scale nested pattern of canopy chemical assembly among thousands of trees. This nested geographic and phylogenetic pattern within and among forest communities **provides a different perspective on current and future** alterations to the functioning of western Amazonian forests resulting from land use and climate change.

Abstract

Patterns of tropical forest functional diversity **express** processes of ecological assembly at multiple geographic scales and **aid in predicting** ecological responses to environmental change. Tree canopy chemistry **underpins** forest functional diversity, **but** the interactive role of phylogeny and environment in determining the chemical traits of tropical trees **is poorly known**. **Collecting and analyzing** foliage in 2,420 canopy tree species across 19 forests in the western Amazon, **we discovered** (*i*) systematic, community-scale shifts in average canopy chemical traits along gradients of elevation and soil fertility; (*ii*) strong phylogenetic partitioning of structural and defense chemicals within communities independent of variation in environmental conditions; and (*iii*) strong environmental control on foliar phosphorus and calcium, the two rock-derived elements limiting CO_2 uptake in tropical forests. **These findings indicate that** the chemical diversity of western Amazonian forests **occurs** in a regionally nested mosaic driven by long-term chemical trait adjustment of communities to large-scale environmental filters, particularly soils and climate, **and is supported by** phylogenetic divergence of traits essential to foliar survival under varying environmental conditions. Geographically nested patterns of forest canopy chemical traits **will play a role in determining** the response and functional rearrangement of western Amazonian ecosystems to changing land use and climate.

Transportation Research Part F: Traffic Psychology and Behaviour
Volume 114, October 2025, Pages 86-98



https://doi.org/10.1016/j.trf.2025.05.023

Never being aggressive is important: A nationally representative survey of aggressive driving in Australia

A.N. Stephens, S.N. How, R. Crotty, J. Oxley

Highlights

- Driving Anger expression inventory was invariant across age, gender and driving purpose.
- Men reported more frequent aggression than women.
- Younger drivers reported more frequent aggression than older drivers.
- Drivers aged 25 to 39 reported more frequent aggression compared to other aged drivers.
- Aggression was associated with increased odds of traffic fines and crashes.

Abstract

The Driving Anger Expression Inventory (DAX; Deffenbacher et al., 2002) is an established tool to measure the frequency of different types of aggressive driving (verbal, personal physical and aggressive use of the vehicle) as well as constructive responses to anger. This study assessed the applicability of DAX on a representative sample of 2,108 drivers from Australia (men = 49.8 %, women = 49.7 %, non-binary/gender-diverse = 0.05 %), ranging in age from 18 to 95 years (M = 46.5; SD = 17.8). Drivers completed an online survey comprising demographic information and the 15-item DAX. Importantly, the scale used in this study sought frequency responses across 5 categories (never, almost never, occasionally, often or almost always); whereas previous applications have not used a never response. Confirmatory Factor Analysis (CFA) showed that the four-factor structure was suitable for drivers in Australia. Multigroup CFA confirmed DAX was invariant across age, gender and main purpose of driving (work or leisure). Aggression was higher for men compared to women; work drivers compared to leisure drivers; and, drivers aged 26 to 39 years compared to younger (aged 18 to 25 years) or older drivers (aged 40 to 64 and 65 + years). After controlling for age and annual kilometres driven, higher scores on DAX were associated with increased odds of having been in a crash or having received a traffic fine in the past 12 months. The findings demonstrate that DAX is a viable tool to measure aggressive driving in Australia which should be administered with a never option.

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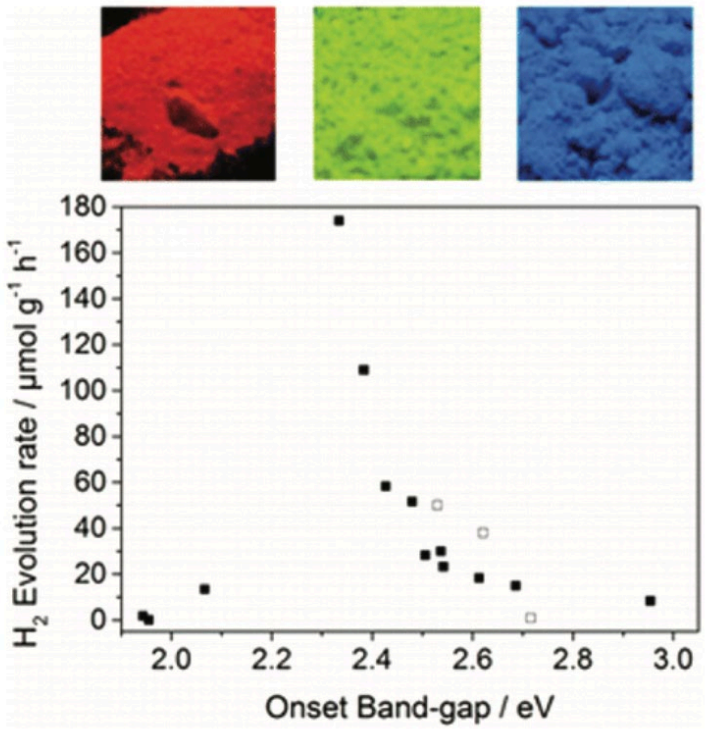
Graphical abstract

- Usually 1 panel (can be composed from several images)
- Can be accompanied by short text
- Image can be taken from the result *or* produce specifically for the abstract
- Provide a clear and memorable image

Graphical abstract

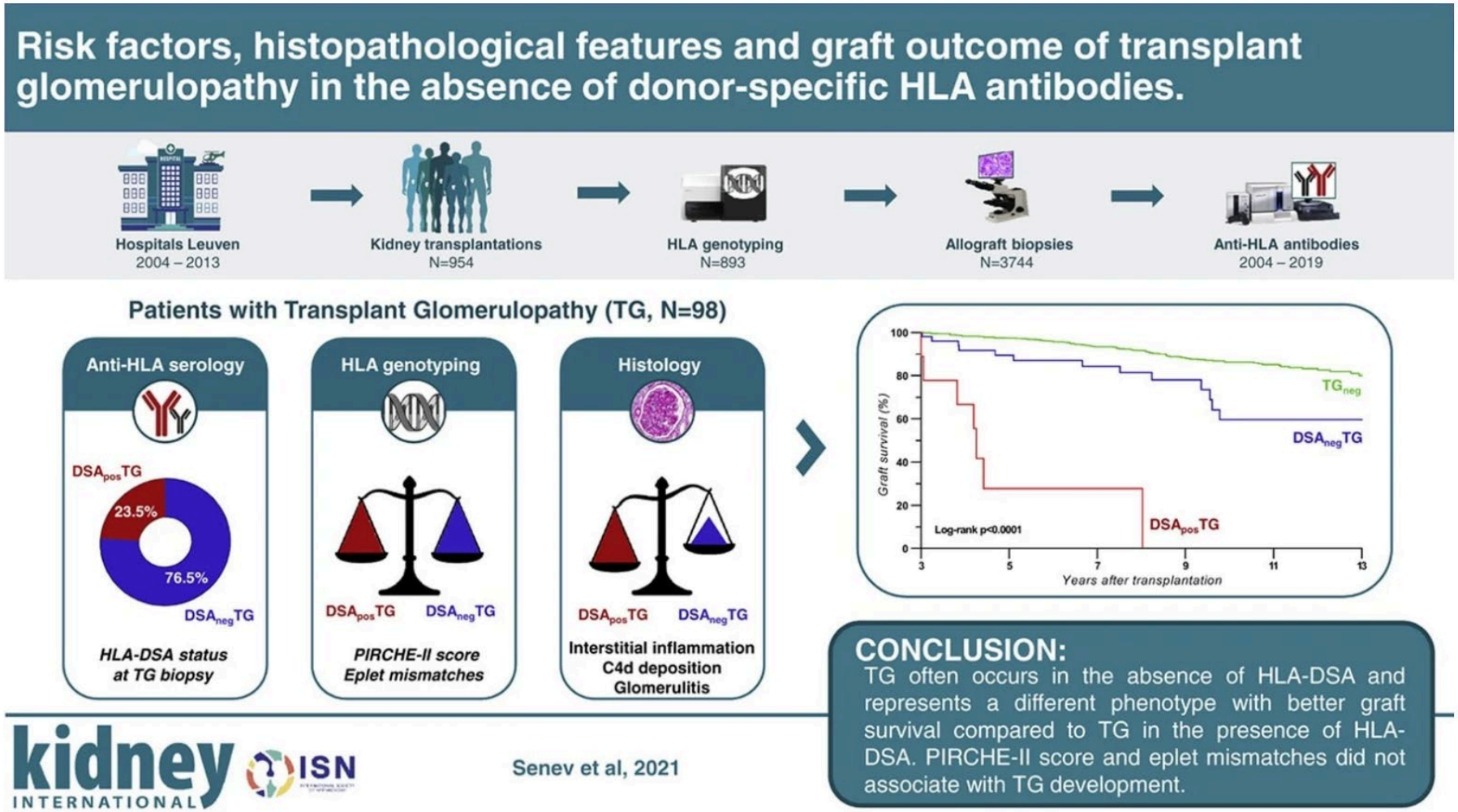
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Tunable organic photocatalysts for visible-light-driven hydrogen evolution



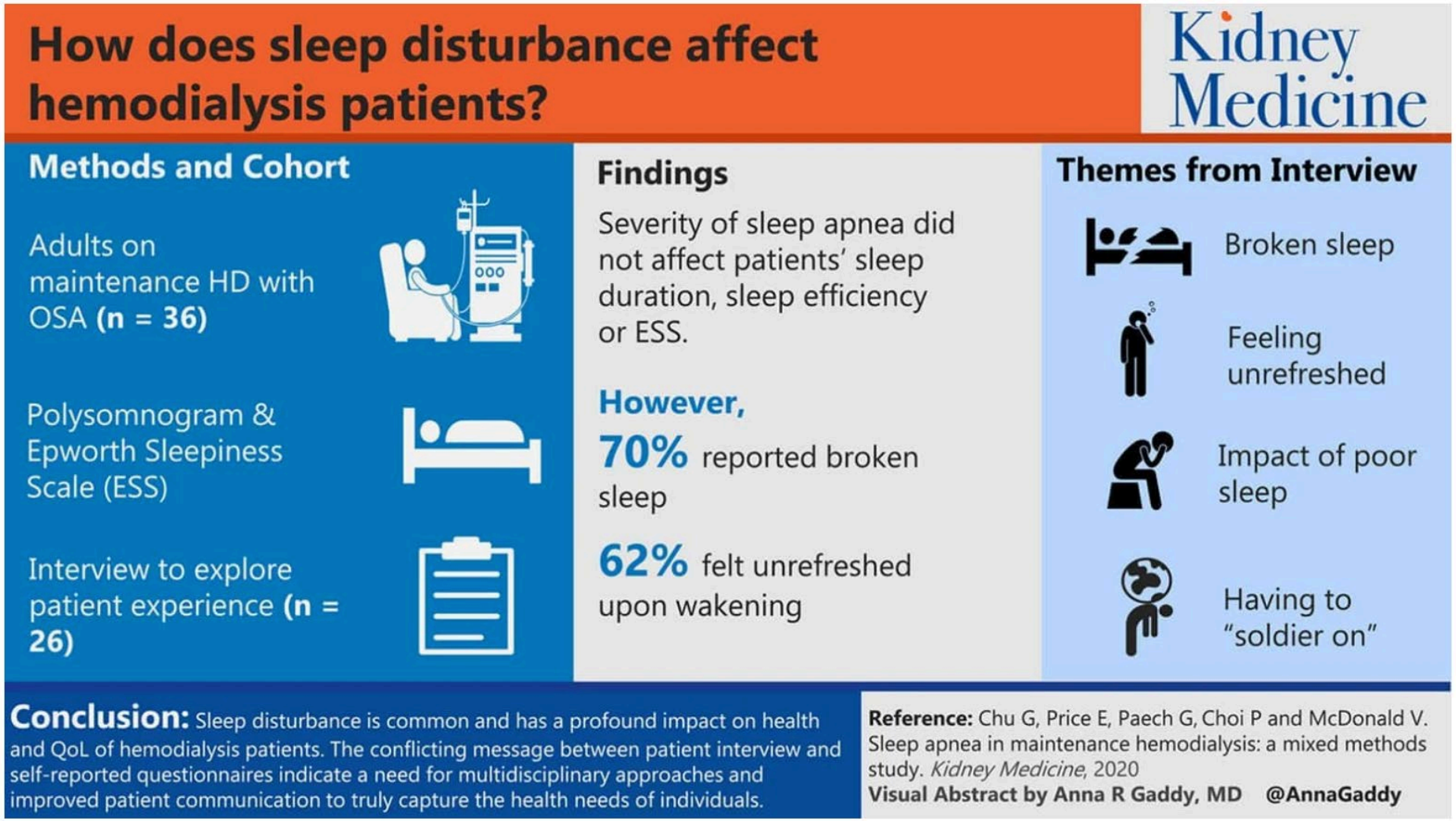
Abstract

Photocatalytic hydrogen production from water **offers an abundant, clean** fuel source, **but it is challenging to produce** photocatalysts that use the solar spectrum **effectively**. Many hydrogen-evolving photocatalysts **are** active in the ultraviolet range, **but** ultraviolet light **accounts for only** 3% of the energy available in the solar spectrum at ground level. Solid-state crystalline photocatalysts have light absorption profiles that are a discrete function of their crystalline phase and that are not always tunable. **Here, we prepare** a series of amorphous, microporous organic polymers **with exquisite synthetic** control over the optical gap in the range 1.94–2.95 eV. Specific monomer compositions give polymers that are **robust and effective** photocatalysts for the evolution of hydrogen from water in the presence of a sacrificial electron donor, **without the apparent need for** an added metal cocatalyst. **Remarkably, unlike other** organic systems, the best performing polymer is only photoactive under visible rather than ultraviolet irradiation.



Risk factors, histopathological features, and graft outcome of transplant glomerulopathy in the absence of donor-specific HLA antibodies ↗

Source: https://www.elsevier.com/researcher/author/tools-and-resources/graphical-abstract



Sleep Apnea in Maintenance Hemodialysis: A Mixed-Methods Study ↗

Source: https://www.elsevier.com/researcher/author/tools-and-resources/graphical-abstract

Generic abstract model

	GENERIC ABSTRACT MODEL
1	SIGNIFICANCE OF THE TOPIC/ESSENTIAL FACTUAL BACKGROUND THE CHALLENGE/PROBLEM WHAT THE PAPER/STUDY DOES (may include ACHIEVEMENT/VALUE + ☺)
2	METHOD/MATERIALS + ☺ RESULTS + ☺/COMPARISONS WITH EXISTING RESULTS + ☺ IMPLICATIONS
3	MAPPING TO EXISTING KNOWLEDGE + ☺ ACHIEVEMENT/VALUE/CONTRIBUTION + ☺ APPLICATIONS + ☺

- Clear key information → *you are writing an article, not a fictional novel, so no suspense please*
- Consider these tenses:
  - What the study is about *or* does → simple present
  - Method → simple past
  - Results and implications → simple present or simple past
  - Achievement or contribution → present perfect or simple present
- Do NOT overclaim

How to write concisely: Combine sentence functions

WHAT THE PAPER DOES + ACHIEVEMENT: *In this paper, we present a novel estimation method that combines the strengths of quality-guided tracking, multi-level correlation, and phase-zero search to achieve high levels of accuracy and robustness.*

ACHIEVEMENT + GAP: *It thereby provides a concept for low-frequency hearing that accounts for a variety of unexplained experimental observations from the cochlear apex, including…*

METHOD + RESULT: *Specifically, addition of ≥ 13.3 mg/ml of 45S5 bioactive glass or Sr50, or ≥ 6.7 mg/ml of Sr100, resulted in significant inhibition.*

ACHIEVEMENT + GAP: *In addition, the present model further improves upon the accuracy of existing models by incorporating a more complete version of the Nernst equation, which enables accurate prediction of the cell potential without the use of a fitting voltage.*

RESULT + IMPLICATION: *In animals lacking functional sensory neurons (TRPV1$^{-/-}$), BMP2-mediated increases in SP and CGRP were suppressed as compared to the normal animals, and HO was dramatically inhibited in these deficient mice, suggesting that neuroinflammation plays a functional role.*

Questions?

Title

- Entry point for prospective readers
- Indicate the topic and scope of the study
- Self-explanatory
- Accurate, brief, and clear
- Working title vs. 'catchy' title → compare:
  - Mapping single-cell gene expression in the lung
  - 'LungGENS: a web-based tool for mapping single-cell gene expression in the developing lung'
- Some journals may use a sentence indicating their most significant findings as the title, e.g., *Gamma-range synchronization of fast-spiking interneurons enhances detection of tactile stimuli*

- Check the title style of your target journal

Compare titles in your journal target

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- Check average length
- Check the use of acronyms
- Title should not be the same with **keywords**
  - Title can be more general while keywords are more specific → attract specialist researcher, OR
  - Title is more specific while keywords are more general → attract wider audiences
- Check the grammar of the title
  - Use 'nouns and compound nouns' wisely
  - Avoid overloading prepositions
- Title model
  - Is it common to use 'a/an' to communicate new offer?
  - The us of a colon (:)
  - Method is included in the title?
  - Is it common to put 'purpose or applications' in the title?
  - The use of happy words?

Evaluating a title

- Does the title predict the focus and content of the article?
- How many nouns are there in each title?
- Is the title easy for all targeted readers to understand?
- Is the title grammatically ambiguous?
- Is the contribution or potential application of the study clear from the title?

Working title or 'catchy' title?

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Abstract

Titles of scientific papers play a key role in their discovery, and "good" titles engage and recruit readers. Humour is a particularly interesting aspect of title construction, but little is known about whether funny titles boost or limit paper impact. We used a panel of scorers to assess title humour for 2439 papers in ecology and evolution, and measured associations between humour and subsequent citation (self-citation and citation by others). Papers with funnier titles were cited less, but this appears to reflect confounding with paper importance: self-citation data suggest that authors give funnier titles to papers they consider less important. After correction for this, papers with funny titles have significantly *higher* citation rates ($P < 2.2 \times 10^{-16}$; roughly doubling from lowest to highest humour score)—suggesting that humour recruits readers. We also examined associations between citation rates and other features of titles. Inclusion of acronyms and taxonomic names was associated with lower citation rates, while assertive-statement phrasing and presence of colons, question marks, and political regions were associated with somewhat higher citation rates. Title length had no effect on citation. Our results suggest that scientists can use creativity with titles without having their work condemned to obscurity.

Heard, et al., 2023. If this title is funny, will you cite me? Citation impacts of humour and other features of article titles in ecology and evolution. *FACETS* 8(1), https://doi.org/10.1139/facets-2022-0079

Title from ‘working title’

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- Bridge reliability assessment based on monitoring
- Influence of cracks on chloride ingress into concrete
- Adsorption energy engineering of nickel oxide hybrid nanosheets for high areal capacity flexible lithium-ion batteries
- Reaction cross sections of carbon isotopes incident on a proton
- Fuzzy analytical hierarchy process for evaluating and selecting a vendor in a supply chain model
- Activation of sulfite by single-atom Fe deposited graphitic carbon nitride for diclofenac removal: The synergetic effect of transition metal and photocatalysis

Cytometry PART A
Journal of Quantitative Cell Science
ISAC INTERNATIONAL SOCIETY FOR ADVANCEMENT OF CYTOMETRY

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Review Article | Free Access

Die hard: Are cancer stem cells the Bruce Willises of tumor biology?

Ákos Fábián, Márk Barok, György Vereb, János Szöllősi

Abstract

In recent years, an exponentially growing number of studies have focused on identifying cancer stem cells (CSC) in human malignancies. The rare CSCs could be crucial in controlling and curing cancer: through asymmetric division CSCs supposedly drive tumor growth and evade therapy with the help of traits shared with normal stem cells such as quiescence, self-renewal ability, and multidrug resistance pump activity. Here, we give a brief overview of techniques used to confirm the stem cell-like behavior of putative CSCs and discuss markers and methods for identifying, isolating, and culturing them. We touch on the limitations of each marker and why the combined use of CSC markers, in vitro and in vivo assays may still fail to identify all relevant CSC populations. Finally, the various experimental findings supporting and contradicting the CSC hypothesis are summarized. The large number of tumor types thus far with a subpopulation of uniquely tumorigenic and therapy resistant cells suggests that despite the unanswered questions and inconsistencies, the CSC hypothesis has a legitimate role to play in tumor biology. At the same time, experimental evidence supporting the established alternative theory of clonal evolution can be found as well. Therefore, a model that describes cancer initiation and progression should combine elements of clonal evolution and CSC theory. © 2008 International Society for Advancement of Cytometry

Catchy title

WIREs WATER

Advanced Review

Talking shit: is Community-Led Total Sanitation a radical and revolutionary approach to sanitation?

Mary Galvin

First published: 22 December 2014 | https://doi.org/10.1002/wat2.1055 | Citations: 36

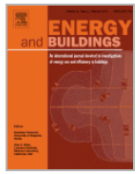
Abstract

Community Led Total Sanitation (CLTS) is a new approach to sanitation that has been widely adopted by international and national development organizations and national governments and implemented in 56 countries in the global South. Using participatory methods, it forces primarily rural communities to recognize that their practice of open defecation causes sickness and disease in their area and 'triggers' them to take action, ensuring that every household builds at least a pit latrine so that the community becomes open defecation free (ODF). In contrast to past approaches, one of its main tenets is strictly no subsidies of finance or materials. In the absence of monitoring and evaluation systems, it is not clear whether its immediate achievements are sustainable. In addition to questioning its sustainability, it is essential to examine CLTS through the analytical lens of power dynamics and human rights. While there is a rich practitioner-focused literature, there are few critical studies of this nature. Drawing on literature from a range of disciplines, this article deliberates how CLTS can be understood in terms of the concepts of rights, agents, and community. It questions whether, in the case of conflicting rights, the communal right to sanitation may justify compromising an individual's right to dignity. It also asks how we balance the right to dignity against the socioeconomic right to sanitation. Finally it questions the community led nature of CLTS, and suggests that external agents retain a level of responsibility for responding to any human rights infringements. *WIREs Water* 2015, 2:9–20. doi: 10.1002/wat2.1055

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Catchy title

Energy and Buildings
Volume 42, Issue 2, February 2010, Pages 173-177



The dark side of occupants' behaviour on building energy use

O.T. Masoso, L.J. Grobler

Abstract

While most studies focus on energy savings during occupied hours, this paper shows the shocking quantities of energy wasted during non-occupied hours in commercial buildings. At least five detailed energy audits were carried out in the hot and dry climates of Botswana and South Africa. The work shows that more energy is used during non-working hours (56%) than during working hours (44%). This arises largely from occupants' behaviour of leaving lights and equipment on at the end of the day, and partly due to poor zoning and controls. There is a crying need for building occupants to learn to switch off what they do not use. The golden rule is: *"If you don't need it, don't use it!"* This is the simplest and cheapest lesson with amongst the biggest savings. Apart from the above, the work also contributes to several other fields of scientific research: it helps in development of benchmarks from sub-hourly field data; it contributes apportionment of energy amongst sub-systems of HVAC, lighting and office equipment; it provides a picture from cooling dominated climates, which normally differs from the largely researched heating dominated climates; it contributes to development of diversity profiles necessary for improvement of simulation accuracy.

Catchy title

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Fifty Shades of Brain: A Review on the Mechanical Testing and Modeling of Brain Tissue

Silvia Budday[1] · Timothy C. Ovaert[2] · Gerhard A. Holzapfel[3,4] · Paul Steinmann[1,5] · Ellen Kuhl[6]

Abstract

Brain tissue is not only one of the most important but also the most complex and compliant tissue in the human body. While long underestimated, increasing evidence confirms that mechanics plays a critical role in modulating brain function and dysfunction. Computational simulations–based on the field equations of nonlinear continuum mechanics–can provide important insights into the underlying mechanisms of brain injury and disease that go beyond the possibilities of traditional diagnostic tools. Realistic numerical predictions, however, require mechanical models that are capable of capturing the complex and unique characteristics of this ultrasoft, heterogeneous, and active tissue. In recent years, contradictory experimental results have caused confusion and hindered rapid progress. In this review, we carefully assess the challenges associated with brain tissue testing and modeling, and work out the most important characteristics of brain tissue behavior on different length and time scales. Depending on the application of interest, we propose appropriate mechanical modeling approaches that are as complex as necessary but as simple as possible. This comprehensive review will, on the one hand, stimulate the design of new experiments and, on the other hand, guide the selection of appropriate constitutive models for specific applications. Mechanical models that capture the complex behavior of nervous tissues and are accurately calibrated with reliable and comprehensive experimental data are key to performing reliable predictive simulations. Ultimately, mathematical modeling and computational simulations of the brain are useful for both biomedical and clinical communities, and cover a wide range of applications ranging from predicting disease progression and estimating injury risk to planning surgical procedures.

Catchy title

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May the Force Be with You: Ultrasound Haptic Feedback for Mid-Air Gesture Interaction in Cars

Authors: Gözel Shakeri, John H. Williamson, Stephen Brewster | Authors Info & Claims

Abstract

The use of ultrasound haptic feedback for mid-air gestures in cars has been proposed to provide a sense of control over the user's intended actions and to add touch to a touchless interaction. However, the impact of ultrasound feedback to the gesturing hand regarding lane deviation, eyes-off-the-road time (EORT) and perceived mental demand has not yet been measured. This paper investigates the impact of uni- and multimodal presentation of ultrasound feedback on the primary driving task and the secondary gesturing task in a simulated driving environment. The multimodal combinations of ultrasound included visual, auditory, and peripheral lights. We found that ultrasound feedback presented unimodally and bimodally resulted in significantly less EORT compared to visual feedback. Our results suggest that multimodal ultrasound feedback for mid-air interaction decreases EORT whilst not compromising driving performance nor mental demand and thus can increase safety while driving.

Catchy title

Discussions?

- What makes you want to read a scientific article?
- Can you understand the article by reading the title and abstract?
- What do you think is a good title?
- Can you identify key messages from reading the abstract?
- As fellow reader, what turn you off the most when reading the title and abstract?

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Thank you